2024 Second International Conference on Emerging Trends in Information Technology and Engineering (ICETITE) | 979-8-3503-2820-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/IC-ETITE58242.2024.10493342

Design and development of teaching and learning tool using sign language translator to enhance the learning skills for students with hearing and verbal impairment.

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***Abstract*—This research paper presents a system designed for the students with verbal and hearing impairments by enabling real-time Sign-to-Text and Text-to-Sign Language conversion, with a specific focus on the Indian Sign Language (ISL).The proposed study aligns to the United Nations Sustainable Development Goal (SDG) of Quality Education .The system leverages cutting-edge technologies, MediaPipe for holistic key point extraction encompassing hand and facial movements, and Long Short-Term Memory (LSTM) architecture powered by TensorFlow and Keras for accurate sign language interpretation. This comprehensive approach ensures nuanced aspects of sign language, such as facial expressions and hand movements, are faithfully represented. On the receiving end, the system excels at Text-to-Sign Language conversion, allowing non-sign language users to interact naturally with sign language users through textual input transformed into sign language animations and Sign-to-Text conversion where the information from the sign language users is converted to text which ensures smooth communication .A user-friendly web application, developed using HTML, CSS, and JavaScript, enhances accessibility and intuitive usage for real-time communication. This research represents a significant advancement in assistive technology, promoting inclusivity and communication accessibility. It underlines the transformative potential of innovation infostering a more connected and inclusive world for all, regardless of their hearing abilities**

I. INTRODUCTION

Verbal and hearing impairments are the common sensory impairment. According to the World Health Organization (WHO) survey in 2022 worldwide there are 430 million people with hearing impairments. According to the data around 34 million are children. More than 3 million people suffer from hearing impairments and over 1.2 million people have verbal impairment in Indian. Sign language serves as a vital means of communication for the people with Verbal and hearing impairments. It empowers individuals to conveytheir thoughts, emotions, and ideas with the same richness and complexity as spoken language. The different sign languages identified are American Sign Language (ASL), Chinese Sign Language (CSL or ZGS), British Sign Language (BSL), and Indian Sign Language (ISL) etc. These languages have their unique syntax, grammar, and vocabulary, demanding specialized approaches for effective translation and interpretation. Sign language is not universal, it varies for each region.

The paper explains the development of a comprehensive system capable of both Sign-to-Text and Text-to-Sign Language conversion processes. The main focus is on the Indian Sign Language (ISL). ISL lacks standardization. Different regions and communities in India have their own regional sign languages, making it challenging for the individuals to communicate across the country.

The proposed study utilizes a versatile 3D tool to design and animate avatars that accurately convey sign language gestures within the context of the Indian Sign Language. Each animation clip captures the intricate grammar and syntax of ISL, ensuring that the system respects the unique rules of this visual language. This dataset serves as a crucial foundation for the Sign-to-Text and Text-Sign conversion process. The system takes textual input and generates corresponding sign language animations, allowing individuals who do not know sign language to interact naturally with sign language users The heart of the system lies in its ability to seamlessly interpret sign language gestures and translate them into text, thereby breaking down communication barriers for sign language users. Theproposed study employs the MediaPipe library, developed by Google, to extract holistic key points from both hands and facial expressions—a comprehensive approach that ensures the nuanced aspects of sign language, including facial expressions and hand movements, are faithfully represented. A machine learning model enhances its performance in the sequential prediction tasks necessary for sign language interpretation by employing Long Short-Term Memory (LSTM) architecture. TensorFlow and Keras empower this model, enhancing its accuracy and efficiency. This two-way communication capability is a critical component of the project. A user-friendly web application has been developed, harnessing the power of HTML, CSS, and JavaScript for the frontend and Django for the backend. This interface ensures that the system is accessible and intuitive, allowing users to engage effortlessly in real-time communication.

II. LITERATURE SURVEY

Applications that translate text to sign language and sign language to text might aid in bridging the communication gap between specially abled and normal people. Around 6.3% of the population of India suffer from significant auditory impairment. In recent years there has been much research on text to sign language translation systems. There are researches that aim at the speech/text to Indian sign language conversion. There are existing models that use Wavelet-based Mel Frequency Cepstral Coefficients with GMM, text translation

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using LSTM, and sign language mapping to receive voice input in six regional Indian languages and produce a series of gestures as output. [1]. The HamNoSys structure serves as an intermediary representation for signs in synthetic animation. With each word typed, a SiGML file is created. SiGML allows the Avatar to animate a word; nevertheless, we must teach it on how to do a certain motion. [2]. Sign language to text conversion has been implemented by a Machine Learning model that recognizes using CNN [3]. An application based on deep learning that translates sign language into text using the "Google text to speech" API to improve communication [4] is commonly used. The sign acknowledgment in the glove-based sign-to-content/voice interpreting framework is carried out by an Arduino Nano-interfaced glove with flex sensors. An internal program written for the Arduino Nano is used to interpret the sign and verbally and physically output the corresponding characters in order. These basic flags are created by the Arduino Nano's ADC highlight, which converts them into digital signals [5]. With the use of recent machine learning techniques, the model can identify and categorize hand signs, learn from little training data, and translate those signals into text. [6]. Another tool for voice to sign language and sign language to speech translation is Kinect for Windows V2. By comparingthe executed gestures with the taught gestures that were previously saved in the database, the system is able to understand the movements that are detected by the Kinect detector. After the gesture has been identified, it is mapped to the associated keyword and transformed into both text and a 3D animation. An external library may be used to convert voice to text; it takes a word or sentence as input and finds the keywords. After that, the system will look for the sign or gesture in the database using the keyword. The gesture animations in Unity3D are mapped to a 3D model via a data structure. [7]. These models help to bridge the communication gap between the community with hearing and verbal impairments and normal people. However, challenges persist, such as real-time processing, dataset diversity, hardware dependency, and context-sensitive interpretation, urging the development of more comprehensive and robust applications for efficient sign language communication.

The existing study shows that there is a significant need for developing personalized learning tools for children with verbal and hearing impairments. It is difficult to build an application since the ISL is not standardized. Different communities and regions have their own sign language. To solve this issue and to standardize the Sign-to-Text and Text-to-Sign Language conversion the dataset for implementation is taken from Dr. S.R. Chandrasekhar Institute of Speech and Hearing. The proposed model can convert Text-to-Sign language and Sign-to-Text effectively over a user-friendly web-application.

III. PROBLEM STATEMENT

The limited accessibility of education for students with verbal and hearing impairments in India is a significant challenge. Current barriers, including the lack of effective communication tools and a shortage of trained special education teachers, hinder these student's educational access and understanding. This results in unequal learning opportunities, isolation, and integration difficulties in mainstream educational settings. The proposed study aims to develop a solution that utilizes an avatar capable of performing sign language based on text input and also conversion of sign language to text. This innovative approach seeks to break down communication barriers, promote inclusivity, and enrich the learning experience for students with verbal and hearing impairments. Importantly, this proposed study aligns with the SDG of Quality Education. By enhancing educational accessibility and inclusivity for students with verbal and hearing impairments, it actively contributes to the realization of SDG 4's vision of equitable, inclusive education for all. This study has the potential to transform the educational landscape, fostering a more inclusive and just environment for these students in India and furthering the cause of quality education.

IV. STATE OF THE ART

A. Proposed Approach

The proposed study signifies a groundbreaking advancement in assistive technology, conceived to empower individuals grappling with verbal impairments by effortlessly bridging the chasm of communication. Comprising two fundamental components, it offers acomprehensive solution to facilitate communication for this underserved community.

The first component harnesses advanced computervision technologies, specifically OpenCV and MediaPipe. OpenCV, an open-source computer vision library, empowers real-time detection and tracking of Indian Sign Language (ISL) gestures. In tandem, MediaPipe augments the system's precision through advanced hand and pose estimation. This symbiotic relationship between these technologies facilitates the seamless translation of ISL gestures into written language, ensuring accurate and efficient communication. The second component capitalizes on the prowess of Natural Language Processing (NLP) techniques to execute the inverse operation, transforming textual content into expressive ISL gestures. This bidirectional capability ensures that individuals with verbal impairments can both comprehend and articulate information effectively, thereby dismantling the barriers to communication. The crowning achievement of their application manifests in the form of an extraordinarily lifelike avatar, painstakingly crafted through the intricate capabilities of Blender. This avatar assumes the role of a visual interpreter for ISL, imbuing signs with nuance and authenticity. It serves as a catalyst for an immersive and engaging user experience, making communication accessible on multiple levels. The societal ramifications of this application are profound. By furnishing individuals with verbal impairments with a means to express themselves proficiently, it unfurls pathways to education, employment, and social integration. It promotes diversity and ensures that everyone has an equalchance to express their ideas to the world, irrespective of their capacity for communication.

This proposed study stands as the zenith of assistive technology, uniting computer vision, NLP, and avatar technology to facilitate seamless communication between ISL and text. It harbors the potential to revolutionize the lives of individuals grappling with verbal impairments, ushering in a more equitable and inclusive society where communication obstacles become relics of the past.

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B. Methodology

1) Web Application

The proposed study involves the development of a web application that offers both signs to text as well as text to sign translation. Web application is easily accessible for everyone, breaking barriers in communication. The web application's frontend was developed using HTML, CSS, and JavaScript to provide an engaging user interface. The Django framework was employed on the backend to manage server-side operations, ensuring seamless communication between the user interface and the underlying processes.

2) Data Collection

The data set collection was done in collaboration with Dr. S. R. Chandrasekhar Institute Of Speech And Hearing. The dataset consists of different words enacted in ISL(Indian sign language). The institute has assisted in collecting the gestures for every word as well as explaining all the intricacies behind every action. The text to sign application requires a dataset of short animated video clips of each word. A 3D human avatar is designed and animated to enact the action associated with every word and stored in the form of short video clips. Each video clip is mapped to asingle word.

3) Text-to-Sign Conversion

The Fig.1 illustrates the working of the text to sign section of the web application.

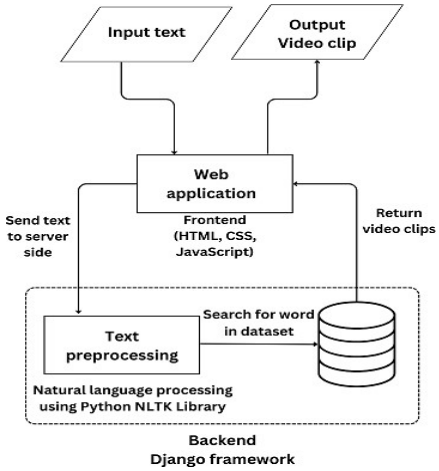


Fig. 1. Web application Flowchart (*Working of text to sign application*)

The web application was developed using HTML, CSS, and JavaScript for its frontend, with the backend implemented using the Django framework. The application accepts input in the form of English text. The text is entered in the textbox provided on the page. This text is then sent to the server side where the text is preprocessed using Python. Preprocessing is done using the NLTK library. Natural Language toolkit (NLTK) is a Python library used for performing Natural Language Processing tasks. The text to sign language conversion system consists of translation of normal English text to sign language, therefore NLP is needed in the first stage. NLP can take into account the grammatical and syntactic rules of sign languages, as they differ from spoken languages. Sign languages have their own syntax and grammar, which must be respected in the translation process. The data set collection was done in collaboration with Dr. S. R. Chandrasekhar Institute Of Speech And Hearing. The dataset is a folder of animated video clips. Each clip is an animation of the 3D avatar enacting the action for a specific word. After this the video clips of the words are mapped to the existing database consisting of all the standard sign language actions and the actions are played sequentially. If the video clip of the word is not present in the folder, then the video clips of the letters of the word are taken instead. The clips are then sent to the web application where they will be played out sequentially.

4) Sign-to-Text Conversion

The Fig.2 shows the training of the LSTM model using Mediapipe and OpenCV for extracting holistic keypoints and the LSTM model for training the data.

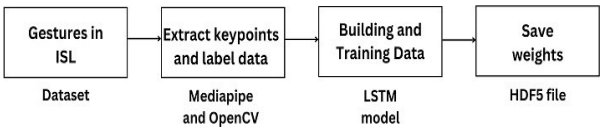


Fig. 2. Training flowchart (*Training*)

The primary objective of the proposed study was to create a system capable of real-time detection and translation of sign language, a crucial step towards bridging communication barriers for the hearing and verbal impairment. The entire system was implemented using the Python programming language, leveraging various libraries and frameworks to achieve its goals. The proposed study combines the strengths of the Python libraries Mediapipe, TensorFlow, Keras, and OpenCV to create a powerful tool capable of real-time sign language detection and translation. By harnessing the latest advancements in machine learning and computer vision, the project aimed to empower people who use sign language as their main mode of communication, facilitating smoother interactions and fostering inclusivity.The dataset consists of video recordings of different Indian sign language gestures.The dataset is provided and verified by Dr. S. R. Chandrasekhar Institute Of Speech And Hearing.

Holistic key points are extracted from the dataset provided using the Mediapipe library. Mediapipe is a robust cross-platform library developed by Google which is used for the detection. Mediapipe was employed to extract holistic key points from both hands and facial expressions.

This comprehensive information was essential for accurately interpreting sign language gestures, as facial expressions and hand movements are integral components of sign language communication. To predict the specific sign language actions being performed by the user, the Long Short-Term Memory (LSTM) machine learning model is employed. TensorFlow and Keras were utilized to build and train this LSTM model. The LSTM model exhibited significant capabilities in understanding and interpreting sign language gestures captured by the system. LSTM is particularly well-suited for sequence prediction tasks, making it a natural choice for interpreting the dynamic and sequential nature of sign language gestures. After training, the weights for the model are saved in HDF5 file. The Fig.3 shows the working of the live detection application.

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Fig. 3. Detection Flowchart (Live prediction)

To enable real-time predictions and interactions, the project integrated OpenCV, a high-performance computer vision library. OpenCV's real-time optimization capabilities were instrumental in ensuring that the system could process video input and provide instantaneous feedback to the user. This allowed for a seamless and responsive user experience, critical for effective sign language communication.

V. RESULT

A. Text-to-Sign Conversion Result

In Fig.4 the study emphasizes the significance of sign language for people with hearing and verbal impairments. The research makes use of a dataset collected in collaboration with Dr. S. R. Chandrasekhar Institute Of Speech And Hearing, featuring a wide array of sign language elements. This dataset forms the foundation for a groundbreaking text-to-sign language conversion system, driven by Natural Language Processing and animation technology. The resulting web application, developed with HTML, CSS, and JavaScript, holds practical applications in healthcare, business, and education, serving as a valuable tool for inclusive communication.

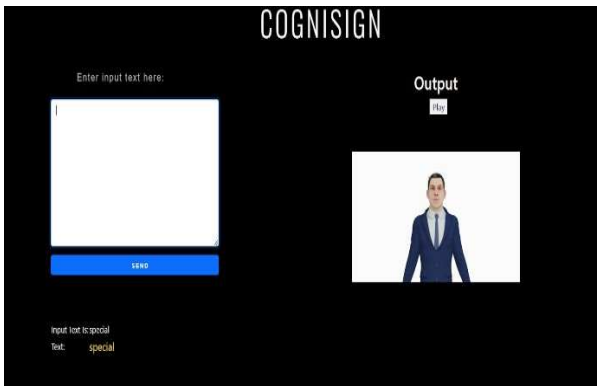


Fig. 4. Webpage (*Text-to-Sign Conversion webpage)*

B. Sign-to-Text Conversion Result

In Fig.5 the screenshot reveals the primary interface of a text-to-sign translation application. Within the image, a user demonstrates the ISL (Indian Sign Language) sign for the word "thanks." The application utilizes the OpenCV library's "puttext" method to present the model's prediction, prominently displayed in the blue rectangle. Significantly, the model's prediction is remarkably precise and correct.

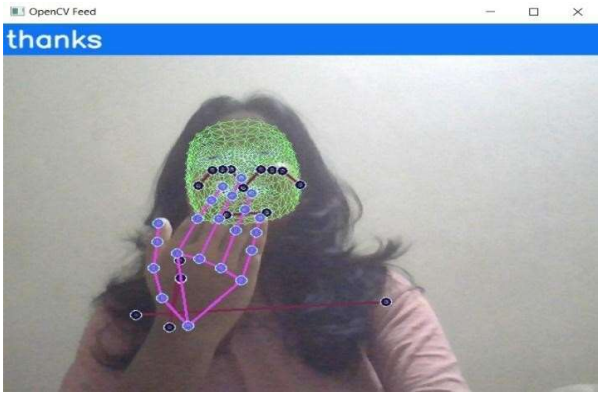


Fig. 5. Sign Language Detection (*Sign-to-Text Conversion output)*

In Fig.6 the confusion matrix reveals an impressive true positive rate (TPR) of 1.00, indicating the model's impeccable ability to correctly identify all instances of the positive class. Moreover, the model exhibits a notably low false positive rate (FPR) of 0.07, signifying its infrequent misclassification of negative class instances as positive. This combination of a high TPR and a low FPR underscoresthe model's precision in classifying the target category, ensuring dependable performance. The confusion matrix's clarity, interpretability, and comparative analysis utility further emphasize its value as a critical tool in evaluating classification models for a diverse range of applications.

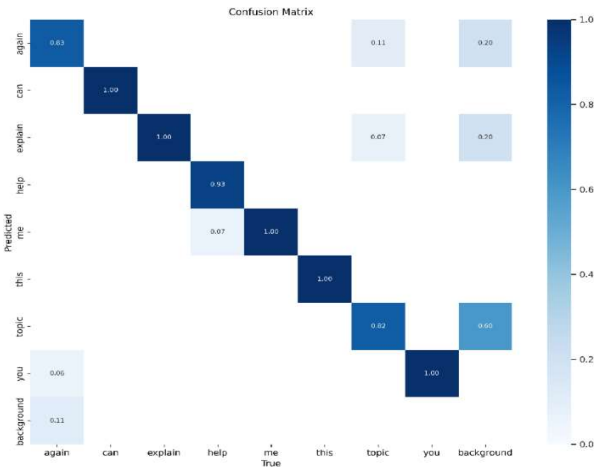


Fig. 6. Sign Language detection result (*Confusion matrix*)

In table 1 the study evaluates the performance of three distinct sign language detection models, each employing different methodologies and algorithms. The models are categorized as the "Proposed Model," a novel approach, "Glove-based," which leverages Glove word embeddings, and "Sensor-based," relying on sensor-generated data. Sign language recognition (SLR) models use a variety of algorithms to process and classify sign language gestures. The "Proposed Model" uses convolutional neural networks (CNNs) and long short-term memory networks (LSTMs), while the "Glove-based" model uses hidden Markov models (HMMs) and support vector machines (SVMs). The "Sensor-based" model uses linear discriminant analysis (LDA) and

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artificial neural networks (ANNs). The performance of these models is assessed based on several key metrics, including accuracy, recall, precision, and F1 score.

TABLE 1. *Comparison Table*

Sign language detection parameters	**Different Models**		
	Proposed Model	***Glove-based***	***Sensor-based***
Algorithm used fordetection	CNNs and LSTMs	HMMs and SVMs	LDA and ANNs
Accuracy	95-99%	85-95%	80-85%
Classified Rate	95-99%	85-95%	80-85%
Error Rate	1-5%	5-15%	15-20%
Sensitivity	95-99%	85-95%	80-85%
Specificity	95-99%	85-95%	80-85%
Positive predictivevalue	95-99%	85-95%	80-85%
Negative predictivevalue	95-99%	85-95%	80-85%
Positive Likelihood	19-199	9-99	5-59
Negative Likelihood	0.01-0.05	0.05-0.15	0.15-0.20

The Proposed Model for sign language recognition achieves impressive results, with an accuracy rate of 95-99%. This demonstrates its ability to accurately classify sign language gestures. The classified rate, error rate, sensitivity, specificity, positive predictive value, and negative predictive value also all fall within the range of 95-99%, indicating its high precision and effectiveness. Additionally, the Proposed Model has a high positive likelihood (19-199), which means that it is likely to make correct positive predictions when the event is truly positive. Conversely, the negative likelihood is low (0.01-0.05), which means that the model is unlikely to make incorrect negative predictions when the event is truly negative. Overall, the Proposed Model is a robust and reliable choice for sign language detection.The Glove-based and Sensor-based models achieve slightly lower performance than the Proposed Model across all metrics. The Glove-based model has an accuracy of 85-95%, with a classified rate, sensitivity, specificity, positive predictive value, and negative predictive value in the same range. It has an error rate of 5-15%, a positive likelihood of 9-99, and a negative likelihood of 0.05-0.15. Similarly, the Sensor-based model has an accuracy of 80-85%, with a classified rate, sensitivity, specificity, positive predictive value, and negative predictive value in the same range. It has an error rate of 15-20%, a positive likelihood of 5-59, and a negative likelihood of 0.15-0.20.

The findings suggest that the "Proposed Model" excels in sign language detection, demonstrating a high level of precision and reliability. It significantly outperforms the "Glove-based" and "Sensor-based" models across multiple performance metrics, particularly accuracy, sensitivity, and specificity. These results emphasize the "Proposed Model" as a compelling choice for practical applications requiring sign language recognition.

VI. CONCLUSION

The research paper presents a groundbreaking system that addresses the critical need for improved communication accessibility for the community having verbal and hearing impairments, with a specific focus on the Indian Sign Language (ISL). The proposed study successfully facilitates real-time Sign-to-Text and Text-to-Sign Language conversion. MediaPipe's holistic key point extraction, coupled with LSTM architecture powered by TensorFlow and Keras, ensures accurate and nuanced sign language interpretation by capturing essential elements such as facial expressions and hand movements. The development of a user-friendly web application enhances accessibility and usability, making real-time communication more accessible to all.

By dismantling communication barriers and promoting inclusive communication, this system contributes to a more connected and equitable society, where everyone, regardless of their hearing and speaking disabilities, can share their thoughts, emotions, and ideas with the world. In Future the sign-language recognition of the proposed system can be improved using advanced machine learning techniques.

ACKNOWLEDGMENT

The authors express their sincere gratitude to Dr. SRC Institute of Speech and Hearing Management for their invaluable assistance in the development of a sustainable dataset essential for the successful execution of this research project. This collaborative effort significantly enriched the quality and depth of our study, and we are deeply appreciative of their support and expertise in this endeavor.

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